System optimizations

At this point, we have a good understanding of:

- Video uploading flow
- Video streaming flow
- Video transcoding architecture

Next, we refine the system using several important optimizations related to:

- Speed
- Safety
- Cost reduction

Speed optimization: Parallelize video uploading

Uploading an entire video as a single file is inefficient, especially for large videos.

To improve performance, the video is split into smaller chunks using **GOP (Group of Pictures) alignment**.

Figure shows how a video is divided into smaller GOP chunks.

Benefits of GOP-based chunking

Faster uploads

Smaller chunks can be uploaded simultaneously.

Resumable uploads

If upload fails:

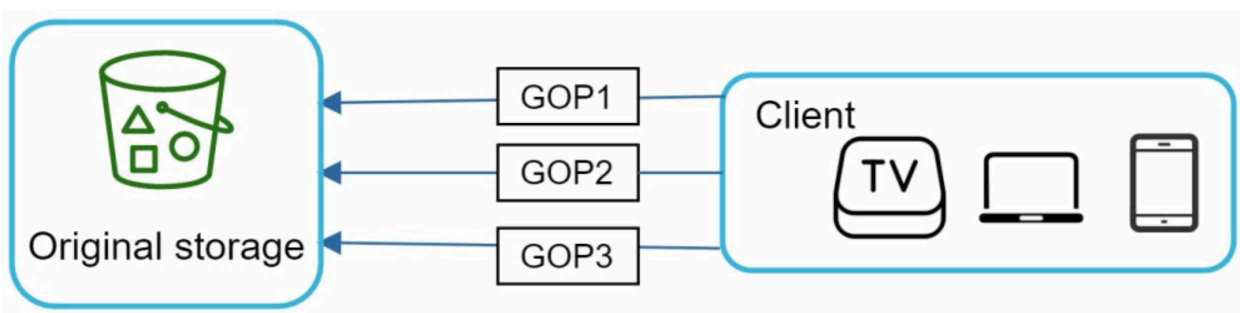
- Only failed chunks are retried
- Entire video does not need to be re-uploaded

Better reliability

Network interruptions affect only a small chunk.

Client-side video splitting

To further improve upload performance, the client itself performs GOP splitting before upload.



This process is shown in **Figure**.

Advantages

Reduced server workload

Servers do not need to split uploaded videos.

Improved parallelism

Multiple chunks upload concurrently.

Faster upload completion

Bandwidth utilization becomes more efficient.

Speed optimization: Place upload centers close to users

Geographical distance increases upload latency.

To reduce latency, multiple upload centers are deployed globally.

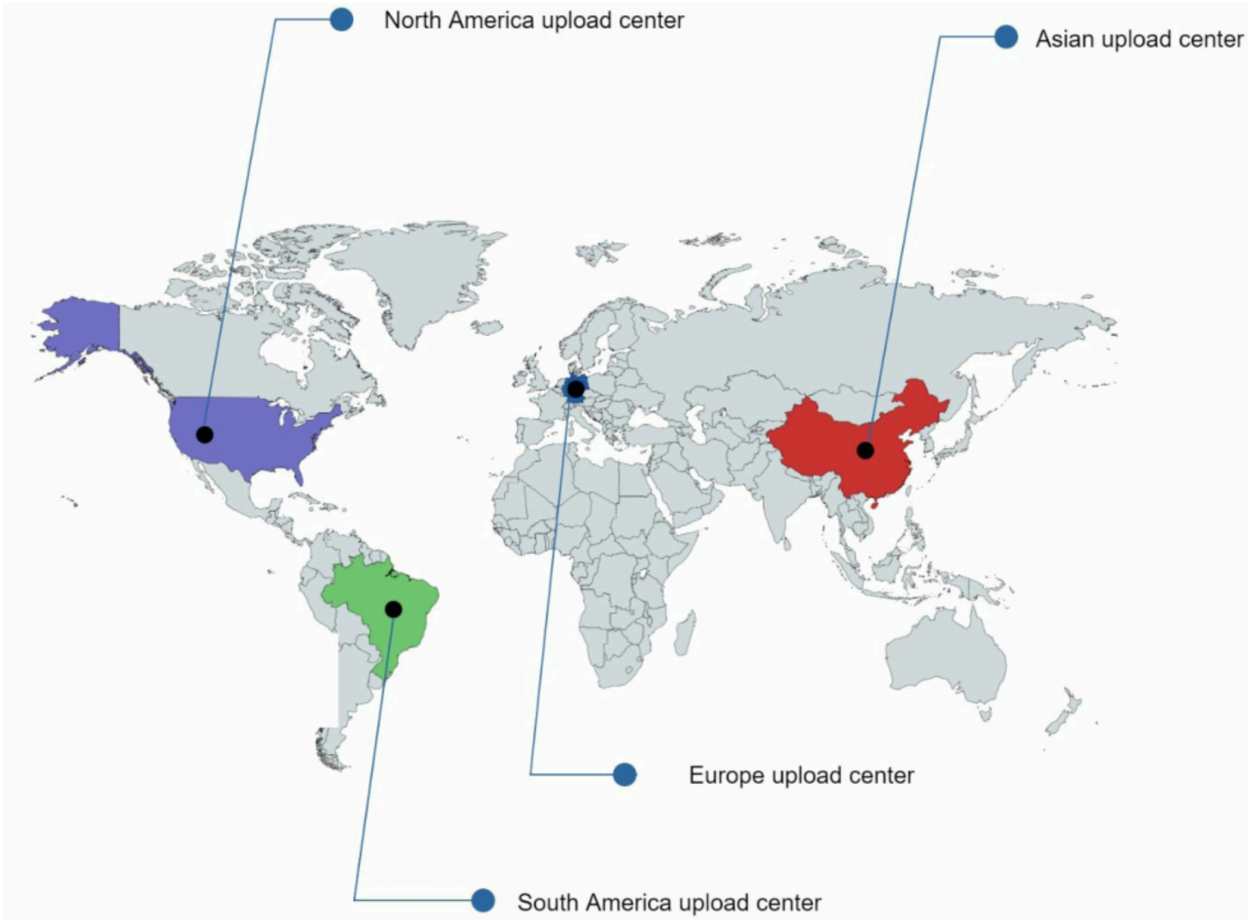


Figure shows globally distributed upload centers.

Examples:

- North America upload center
- Asia upload center
- Europe upload center

Working

Users upload videos to the nearest upload center:

- US users → North America center
- Asian users → Asia center

CDNs can also function as upload centers.

Benefits

Lower latency

Shorter network distance improves upload speed.

Better bandwidth utilization

Regional routing reduces congestion.

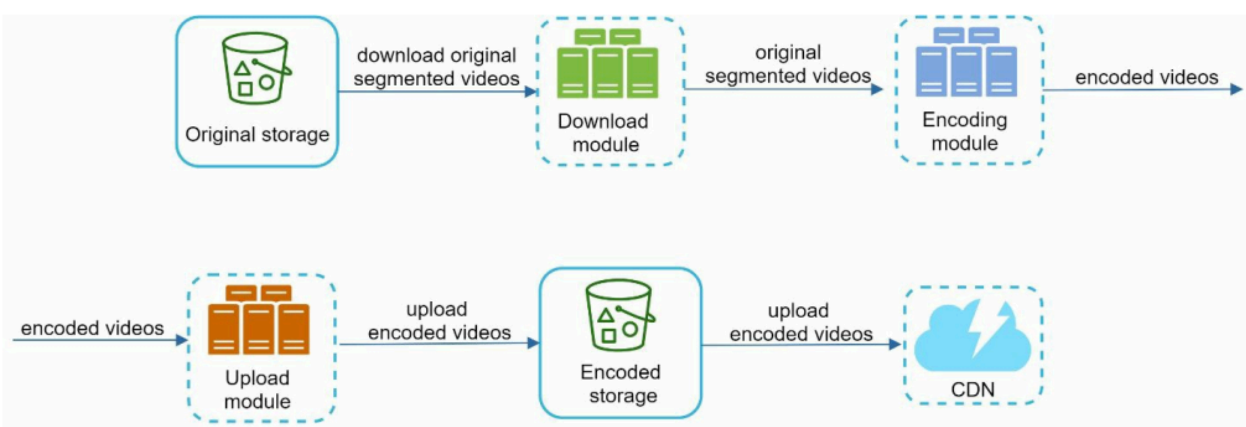
Improved user experience

Uploads become smoother and faster.

Speed optimization: Parallelism everywhere

High-performance systems require maximum parallel execution.

Initially, the workflow from original storage to CDN is tightly coupled.



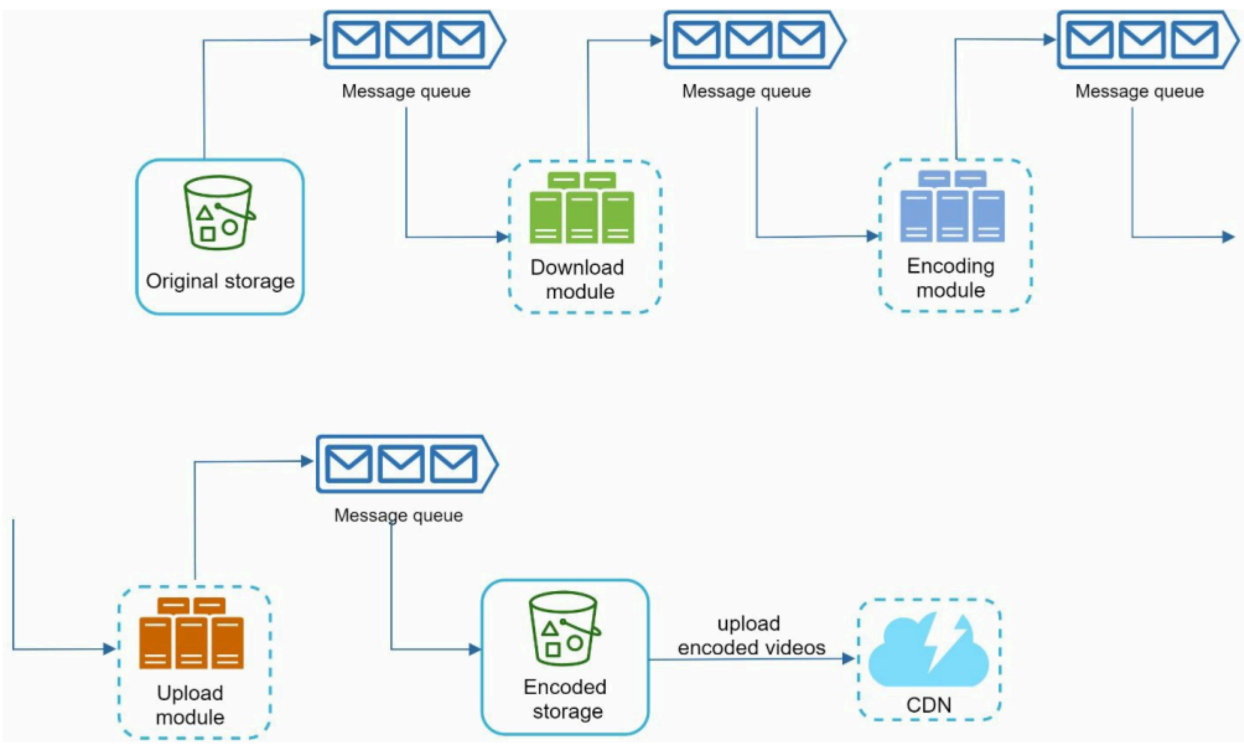
This dependency chain is shown in **Figure**.

Problem:

- Each module waits for the previous module to complete
- Parallelism becomes difficult

Using message queues for loose coupling

To improve scalability and concurrency, message queues are introduced.



This improved architecture is shown in **Figure**.

Before message queues

The encoding module must wait for:

- Download module completion

This creates bottlenecks.

After message queues

Modules communicate asynchronously through queues.

Now:

- Download service pushes events into queue
- Encoding workers consume jobs independently

Benefits:

- Loose coupling
- High parallelism
- Better fault isolation
- Easier scaling

Advantages of message queues

Independent scaling

Each service scales separately.

Fault tolerance

If one module fails, others continue working.

Better throughput

Multiple workers process tasks simultaneously.

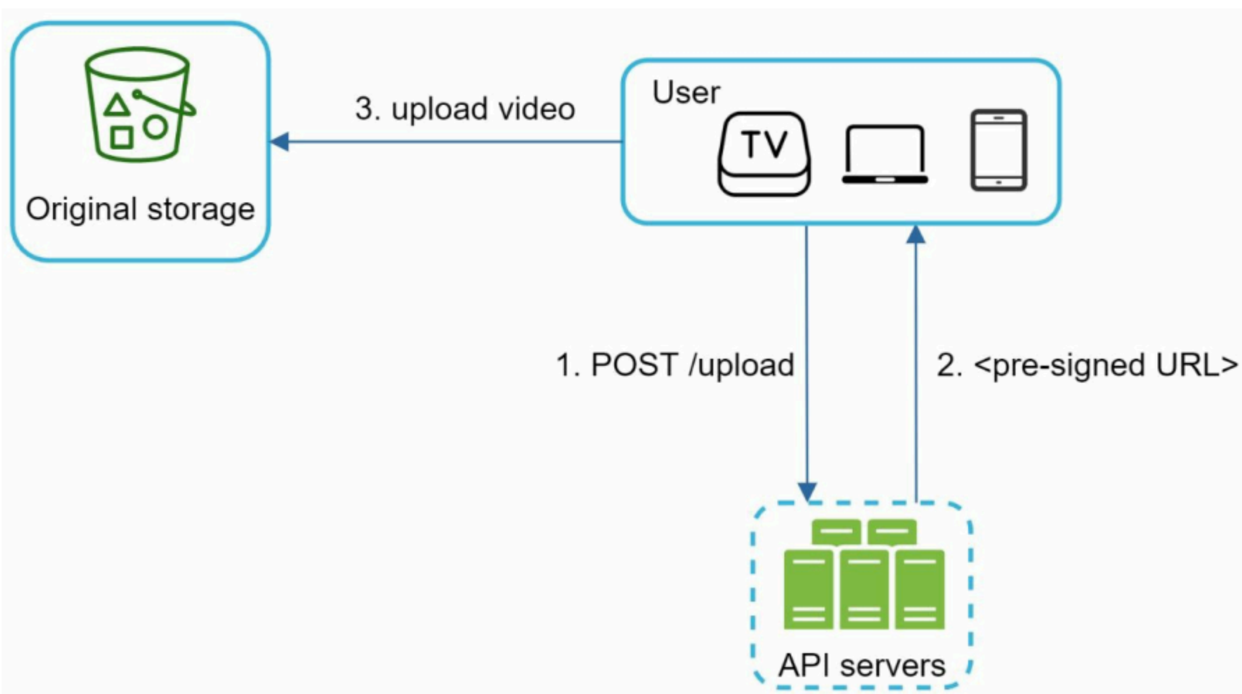
Asynchronous processing

Long-running tasks no longer block workflows.

Safety optimization: Pre-signed upload URL

Security is critical in any large-scale platform.

To ensure that only authorized users upload videos, we use **pre-signed URLs**.



The workflow is shown in **Figure**.

Updated upload flow using pre-signed URLs

Step 1

Client sends an HTTP request to API servers requesting upload permission.

Step 2

API server generates and returns a pre-signed URL.

The URL:

- Grants temporary upload permission
- Specifies upload location
- Expires after a certain time

Step 3

Client uploads the video directly to blob storage using the pre-signed URL.

Benefits of pre-signed URLs

Improved security

Unauthorized users cannot upload files.

Reduced API server load

Video traffic bypasses API servers.

Better scalability

Blob storage handles uploads directly.

Temporary authorization

URLs expire automatically after a short duration.

Safety optimization: Protect your videos

Content creators want protection against:

- Piracy
- Unauthorized downloads
- Content theft

Several protection mechanisms are used.

1. Digital Rights Management (DRM)

DRM systems control:

- Access
- Playback
- Copy protection

Popular DRM systems:

- Apple FairPlay
- Google Widevine
- Microsoft PlayReady

Benefits:

- Strong copyright protection
- Controlled access

2. AES encryption

Videos are encrypted before storage or streaming.

Workflow

- Video is encrypted
- Authorized users receive decryption access
- Video decrypts during playback

Benefits:

- Prevents unauthorized viewing
- Secures premium content

3. Visual watermarking

A watermark is added on top of the video.

Examples:

- Company logo
- User ID
- Brand name

Benefits:

- Discourages piracy
- Identifies leaked content

Cost-saving optimization

CDN delivery is expensive because:

- Video files are large
- Traffic volume is enormous

Back-of-envelope calculations showed huge CDN costs.

To reduce expenses, several optimizations are used.

Long-tail distribution observation

Research shows YouTube traffic follows a **long-tail distribution**:

- Few videos are extremely popular
- Most videos receive little traffic

This observation enables multiple cost optimizations.

1. Serve only popular videos from CDN

This optimization is shown in **Figure 14-28**.

Strategy

- Frequently watched videos → CDN
- Less popular videos → cheaper storage servers

Benefits:

- Reduced CDN bandwidth cost
- Efficient cache utilization

2. Encode less popular videos on-demand

Rarely watched videos do not need all encoded versions pre-generated.

Instead:

- Encode only when requested

Benefits:

- Reduced storage cost
- Lower processing overhead

3. Regional video distribution

Some videos are popular only in certain geographic regions.

Examples:

- Local language videos
- Regional news
- Country-specific content

Strategy:

- Store videos only in relevant regional CDNs

Benefits:

- Reduced replication cost
- Better bandwidth efficiency

4. Build your own CDN

Large streaming companies may eventually build private CDNs.

Examples:

- Netflix Open Connect

The platform can partner with:

- ISPs
- Telecom providers

Examples of ISPs:

- Comcast

- Verizon
- AT&T

Benefits:

- Lower bandwidth charges
- Better streaming performance
- Reduced dependence on third-party CDNs

Importance of analytics

All optimizations rely heavily on:

- Historical viewing patterns
- User access behavior
- Video popularity
- Geographic demand

Analytics helps determine:

- Which videos stay in CDN
- Which videos move to cold storage
- Which resolutions should be generated

Summary of system optimizations

Optimization	Benefit
GOP chunk uploads	Faster resumable uploads
Global upload centers	Lower latency

Message queues	High parallelism
Pre-signed URLs	Secure uploads
DRM & encryption	Copyright protection
CDN optimization	Lower infrastructure cost
Regional distribution	Efficient bandwidth usage
On-demand transcoding	Reduced storage cost